PIC 10A 1A

TA: Bumsu Kim

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Today…

- HW5 Part 2 Hints
- 2-D and 1-D Vector Exercise: “Permutations”

HW5 Part 2 Hints

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- Two different approaches
- Strategy A: Use that $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$
  - Pros: You can reuse my *HW3 Array solution* (on GitHub)
  - Cons: inefficient (=not fast) … but the efficiency doesn't matter for the HW score anyway

- Strategy B: Direct comparison after *sorting* elements
  - Pros: faster, and you can reuse the *array sort function* (on GitHub)
    - (Tuesday's version of the *array sort function* has some error, and it's fixed now)
  - Cons: Needs a special handling for duplicate elements
    - Approach 1: Ignore duplicates while getting user input
      - See *HW3 Array solution* for this!
    - Approach 2: Handle duplicates during the comparison, after sorting
      - See next slides for this

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Strategy A: Use $A = B$ *iff* $A \subseteq B$ and $B \subseteq A$

- First check if $A \subseteq B$:
  - For each member a of A:
    - check if a is in B
  - If every member of A is in B, conclude $A \subseteq B$.
- Secondly, change the roles of A and B and check if $B \subseteq A$

- For the membership check, use "has_member()" function in "Week6_HW3_Array_Sol.cpp"

```
/**
* Checks if the array [arr] has a member [n] or not
* @param arr: array to check
* @param sz: size of the array
* @param n: a data to check
* @return true if and only if arr has n
*/
bool has_member(int arr[], int sz, int n) {
	for (int i = 0; i < sz; ++i) {
		if (arr[i] == n) {
			return true;
		}
	}
	return false;
}
```

```
for ( /* each member a in A */) {
    /* check if B has a */
    // if so, do what?
    // otherwise, what?
}
```

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Strategy B: Direct Comparison After Sorting

- First sort both arrays, then compare the sorted arrays

- Handling duplicates:
  - Approach 1: Remove duplicates when getting input (See HW3_Array_Sol.cpp for this!)

A : 1, 4, 9, 16, 7, 11
B : 11, 7, 9, 16, 4, 1

$\xrightarrow{\text{Sort}}$

A : 1, 4, 7, 9, 11, 16
B : 1, 4, 7, 9, 11, 16

Easy to compare: Use part 1 of this HW!

  - Approach 2: Ignore duplicates after sorting the arrays

A : 1, 4, 9, 16, 9, 7, 4, 9, 11
B : 11, 11, 7, 9, 16, 4, 1

$\xrightarrow{\text{Sort}}$

A : 1, 4, 4, 7, 9, 9, 9, 11, 16
B : 1, 4, 7, 9, 11, 11, 16

Then compare numbers ignoring duplicates

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Comparison of Two Strategies

- Strategy A is pretty straightforward, but note that it is computationally inefficient
  - Let m = #(entries of A) and n = #(entries in B) counting duplicates.
  - We need m membership checks for A, and n checks for B
  - Each requires n or m comparisons in the worst case, so total $m * n$ comparisons

- On the other hand, the sorting method is much more efficient
  - 1. Sort A and B (requires $m \log m + n \log n$ computations – don't need to know why; covered in 10B)
  - 2. Comparison (requires at most $\min(m, n)$ comparisons)

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Summary

- To sum up, [Strategy B] – [Approach 1] is recommended, because you can reuse the following functions/code snippets that we *already* have:
  - read_inputs() [Week7_Array_Functions_sol.cpp]
    - with has_member() [Week6_hw3_array_sol.cpp] for a modification as in [Week6_hw3_array_sol.cpp] to ignore duplicates
  - sort() and its helper functions [Week7_Array_Functions_sol.cpp]
  - Comparison of two arrays [Part 1 of HW5]

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2-D and 1-D Vector Exercise: “Permutations”

Find all permutations of given numbers

Permutations

- Given a sequence of distinct numbers (as a vector<int>), find all permutations and return them as a vector of vectors

- For instance,

Given: {1, 2, 3}
as a 1-D vector

Answer:
{
{1, 2, 3}, {1, 3, 2},
{2, 1, 3}, {2, 3, 1},
{3, 1, 2}, {3, 2, 1},
}
as a 2-D vector
(vector of vectors)

- [Live Coding]

- Note: Solution is already on GitHub!

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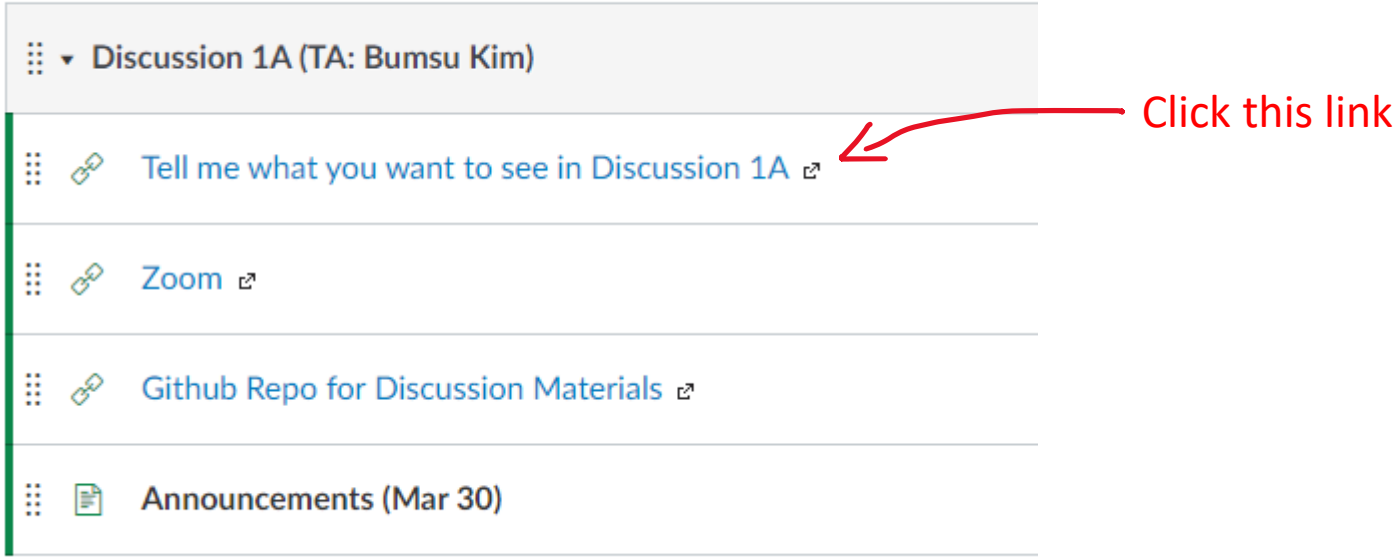
Your Feedback is welcome

- Don’t hesitate to give a feedback on the discussion
- You can use a link in BruinLearn (Google Form)